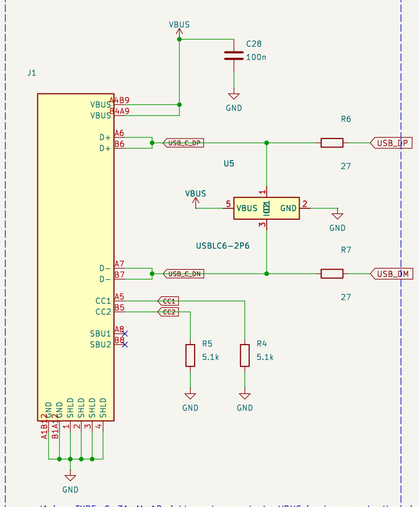
simple_earring — generated schematic

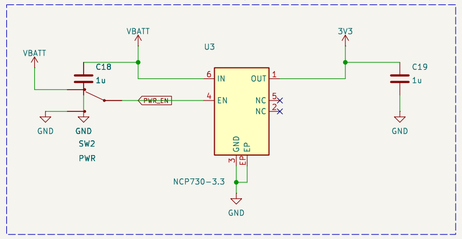
123 components / 81 nets / 4-layer JLCPCB stackup. Power rails (GND, 3V3, VBATT, VLED, VBUS, VDD_CORE) are power symbols; only cross-block signals use global labels. Generated from scripts/circuit.py — do not hand-edit; rebuild with `python3 build.py --only schematic`.

USB-C receptacle / ESD / series R

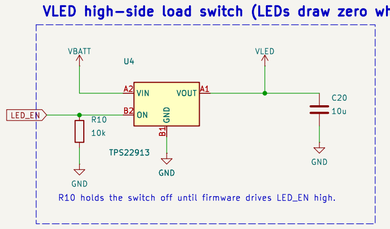


J1 is a TYPE-C-31-M-12 right-angle receptacle. VBUS is charge-only: there is no VBUS->3V3 path, the MCU always runs from the cell.

3V3 LDO + power slide switch (switches EN, not the cell)

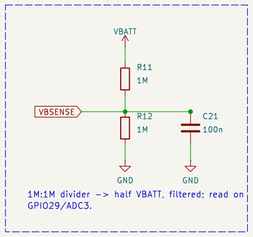


VLED high-side load switch (LEDs draw zero when off)



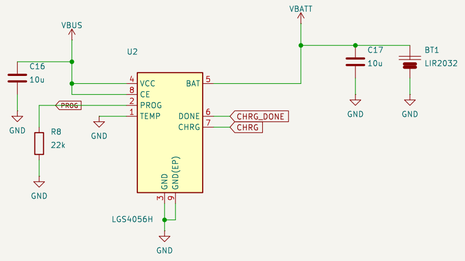
R10 holds the switch off until firmware drives LED_EN high.

Battery voltage sense -> ADC



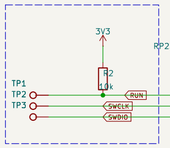
1M:1M divider -> half VBATT, filtered; read on GPIO29/ADC3.

Li-ion charger (LGS4056H, ~41mA) + LIR2032 cell

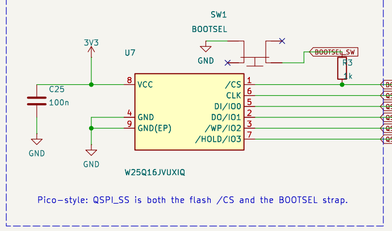


CHRG / DONE are open-drain -> GPIO with internal pull-up (low = active); software owns the charge indicator.

RUN / SWD

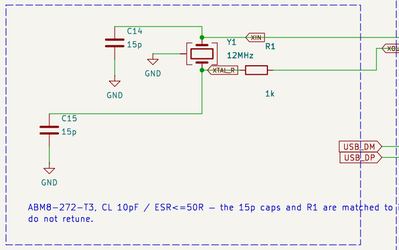


QSPI flash (BOOTSEL strap on /CS)



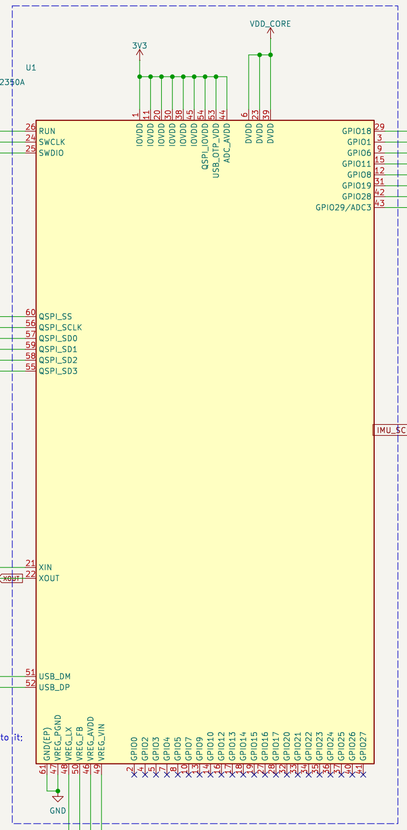
Pico-style: QSPI_SS is both the flash /CS and the BOOTSEL strap.

12MHz crystal

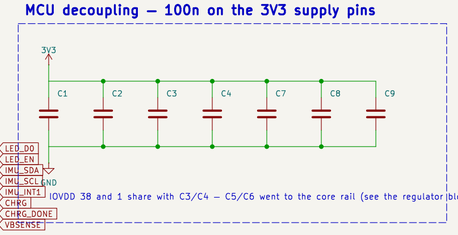


ABM8-272-T3, CL 10pF / ESR<=50R — the 15p caps and R1 are matched to it; do not retune.

RP2350A

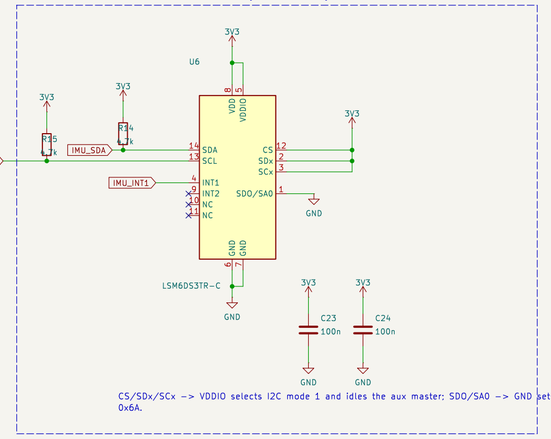


MCU decoupling — 100n on the 3V3 supply pins



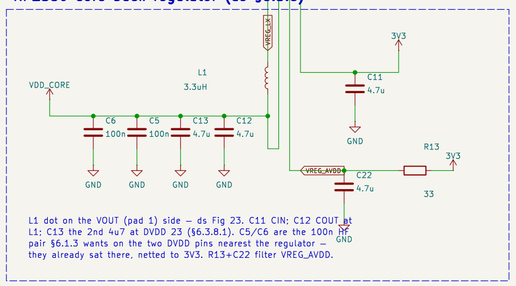
IOVDD 38 and 1 share with C3/C4 — C5/C6 went to the core rail (see the regulator block)

IMU — LSM6DS3TR-C on I2C1 (addr 0x6A)



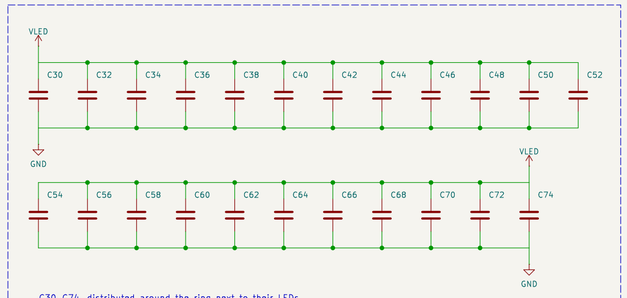
CS/SDx/SCx -> VDDIO selects I2C mode 1 and idles the aux master; SDO/SA0 -> GND sets address 0x6A.

RP2350 core buck regulator (ds §6.3.8)



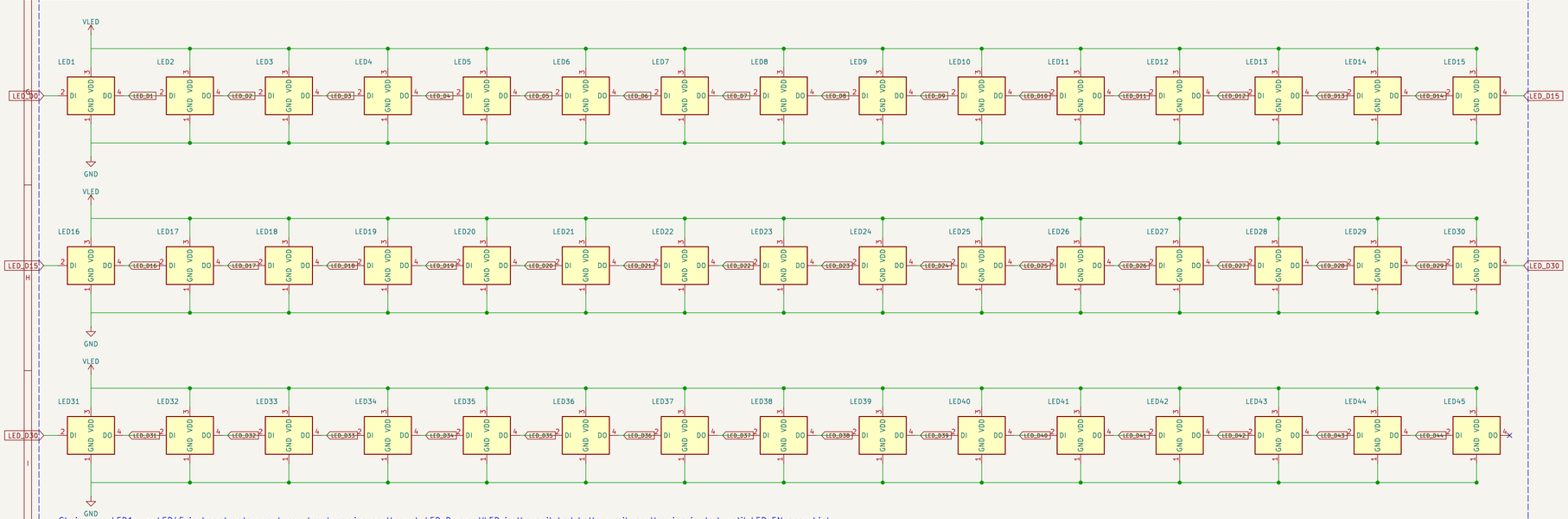
L1 dot on the VOUT (pad 1) side — ds Fig 23. C11 CIN; C12 COUT_bt pair §6.1.3 wants on the two DVDD pins nearest the regulator — they already sat there, netted to 3V3. R13+C22 filter VREG_AVDD.

LED ring bypass — 100n VLED-GND, one per two LEDs



C30..C74, distributed around the ring next to their LEDs

Addressable LED ring — 45 x SK6805-EC10 daisy chain



Chain runs LED1 -> LED45 in board order; each row break carries on through LED_D<n>. VLED is the switched battery rail, so the ring is dark until LED_EN goes high.

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KiCad E.D.A. 9.0.8	Rev: Id: 1/1